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SPECIAL ISSUE -TECHNICAL NOTE

Automated generation of a finite element stent model

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Abstract Numerical simulations have proven to be a valuable tool to investigate the mechanical behavior of stents. These computer models require a considerable amount of preprocessing and computational effort and consequently there is a continuous need for accurate simplifications and automation. For example, it was recently shown that using beam elements instead of solid elements results in a significant speed up of stent simulations. However, the currently applied techniques to create a finite element mesh starting from stent samples remain time-consuming. We present a semi-automated strategy to obtain an accurate finite element beam mesh from a stent sample. The method consists of two steps: (1) A triangulated surface representation of the stent geometry is obtained from micro CT images. (2) Subsequently, a beam mesh is automatically generated by computing the centerline. The method is time-effective and results in an accurate 3D stent model as demonstrated for the MULTI-LINK Vision™ stent.

Keywords Stent · Finite element mesh · Centerline

1 Introduction

Stent implantation has emerged as a better alternative to treat stenosed arteries compared to balloon angioplasty [14]. However, in-stent restenosis remains a major concern when dealing with complex lesions, like stenosis at the location of bifurcations and in the superficial femoral artery [2, 13]. These drawbacks of stenting justify the wide range of proposed in vitro and numerical models to study and improve the existing stent designs [4–7, 9–11]. The applicability of the current numerical models is restricted by the required amount of preprocessing and computational effort, which is significant. As a result, there is a need to reduce the time cost of the complete process, from stent sample to simulation. This can be done by using acceptable model simplifications, such as the one proposed by Hall and Kasper [5]. They have shown that approximating the stent geometries with beam elements yields acceptable results when studying the global stent stress state after expansion and considerably decreases the computational time.

In addition to the computational cost, there is a considerable effort needed to obtain accurate finite element meshes (beam and/or solid) starting from existing stent geometries. The commonly used procedure involves different manual time-consuming steps: (1) determination of the stent dimensions, (2) creation of the 3D CAD model, and (3) mesh generation [4, 9–11] (Fig. 1). Furthermore, this procedure may fail to accurately replicate existing stent designs, because non-uniform deformations resulting from the crimping process (i.e., a fixation of the stent on a folded angioplasty balloon by reducing its diameter) are not taken into account.

In this study, we present a new strategy to create meshed stent models for finite element analysis starting from stent samples in a semi-automated way. The method is based on automatic centerline extraction from a triangulated surface and yields a beam mesh accurately representing the stent geometry. Such a triangulated surface is obtained from

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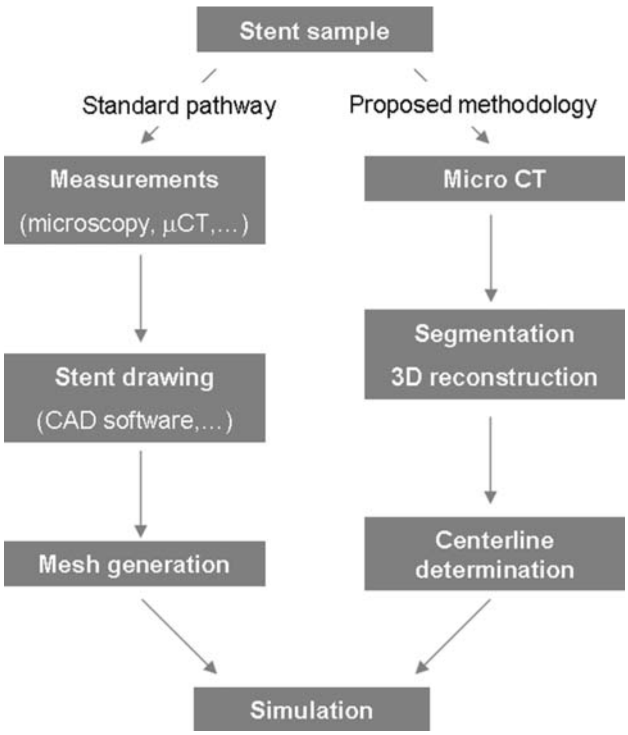


Fig. 1 Two different strategies can be used to obtain a finite element mesh starting from a stent sample. The *left pathway* illustrates the previously used methodology, which involves several manual steps. The different steps of the alternative (semi-automated) strategy presented in this work are shown in the *right pathway*

micro-CT images after segmentation and 3D reconstruction. Micro-CT reconstruction of stent geometries has been used in the past [12]. However, to the author's knowledge, these works were related to computational fluid dynamics and a method to create a suitable mesh for finite element analysis starting from a reconstructed stent has not been proposed yet. The feasibility and time-effectiveness of the proposed methodology is demonstrated for the MULTI-LINK Vision^{TM} stent.

2 Materials and methods

The micro-CT setup used for this investigation consists of a Feinfocus X-ray tube with beryllium exit window, a sample manipulator with air bearing rotation stage and a Varian Paxscan 2520V X-ray flat panel detector [8]. For the measurements we used the X-ray tube at 80 kV and 2.7 W which results in a focal spot size of 3 µm. The detector was used in binning mode with a pixel size of 254 µm. The field of view of the detector is 200 mm (width) by 160 mm (height). The distance between the source and the detector was 880 mm and the distance between the center of the stent and the X-ray source was 36 mm. This geometry results in a magnification of 24 times and a CT voxel size of 10.6 µm. The total scan time was 6 min for 900 projection images.

For this study, one commercially available stent was selected and scanned in its crimped state: the MULTI-LINK Vision^{TM} (nominal diameter 3 mm, nominal length 8 mm, Abbott Vascular). The CT reconstruction was conducted with OCTOPUS software (http://www.xraylab.com). Mimics (Materialise, Leuven, Belgium) was used for the segmentation and 3D reconstruction (http://www.materialise.com/mimics). The software package uses simple gray-value thresholding on the CT-slices in order to obtain a 3D mask of the object of interest. This mask could be considered as a large 3D matrix with Boolean values in each of its element, determining whether the element is part of the virtual object or not. This mask is then used to derive a triangulated mesh which is further processed by triangle reduction and smoothing operations using MAGICS (Materialise, Leuven, Belgium), the integrated remeshing feature of MIMICS.

An approximated centerline was determined starting from the triangulated surface using pyFormex, open-source software currently under development at our institution (http://pyformex.berlios.de). The main steps in the applied centerline algorithm are:

1. Calculation of the Delaunay tessellation [1].
2. Calculation of the embedded Voronoi diagram [1].
3. For each Voronoi vertex (i.e., the circumcenter of a Delaunay tetrahedron), the radius of the corresponding Voronoi sphere is computed.
4. The Voronoi vertex with the largest corresponding radius is added to a new list and all Voronoi vertices within this Voronoi sphere are deleted. This is repeated until all Voronoi vertices are deleted. The result is a subset of the original list of Voronoi vertices.
5. Two Voronoi vertices from the subset are connected if the distance between these two vertices is smaller then the sum of their corresponding radii.

The resulting centerline consists of connected line segments, that can directly serve as a finite element beam mesh. The beam section can be estimated from the triangulated surface. A flow chart of the complete process is depicted in Fig. 1.

The Multi-Link Vision^{TM} stent was virtually expanded using both the standard pathway and the new approach (based on the centerline calculation) in order to verify the accuracy of the new method. The simulations were performed with the ABAQUS finite element solver (Dassault Systèmes). The two beam models were deployed to a 3-mm diameter by increasing the radius of a rigid cylinder [4]. The foreshorting, defined as $(l_0 - l)/l_0$,

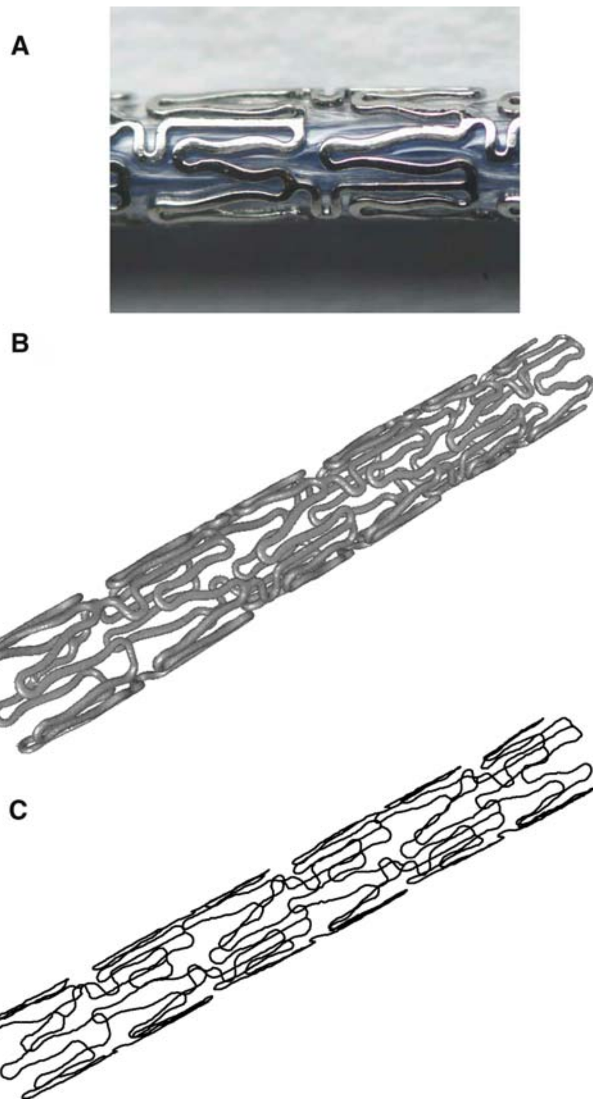


Fig. 2 **a** A photograph of the stent. The surface mesh obtained after segmentation of the CT-images is shown in **b**. The resulting finite element beam mesh obtained by the centerline determination is depicted in **c**

was calculated for both stent models (l_0 is the stent length in crimped state, l is the stent length in expanded state).

3 Results and discussion

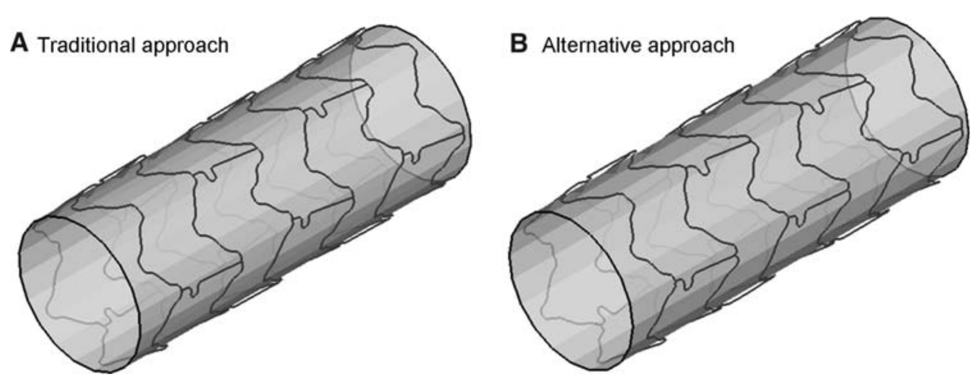
The resulting finite element beam mesh (or centerline) is depicted in Fig. 2 (bottom panel) and was obtained from the surface mesh containing 171,110 triangles within few minutes on a standard desktop computer (2 Ghz processor). The mesh consists of 2,268 beam elements and accurately resembles the MULTI-LINK Vision^{TM} stent geometry. This mesh can be used to simulate the stent expansion and/or to investigate the mechanical behavior of the stent [5, 7].

Figure 3 shows the comparison between two virtually deployed Multi-Link Vision^{TM} stents. The simulation based on the traditional approach (2,046 beam elements) is depicted in panel A, whereas the beam mesh in panel B is based on the new approach involving the centerline computation. The expanded stent geometries are very similar and this is also quantitatively reflected by the foreshortening, which is identical in both cases (i.e., 2.5%). Consequently, the beam mesh obtained by calculating the centerline seems an accurate alternative to a traditionally created beam mesh.

Although the finite element beam mesh is automatically created from a triangulated surface, a certain amount of manual work is still required to obtain this surface mesh starting from a stent sample. This manual work is necessary to prepare the micro-CT setup (positioning of the sample) and to segment the images (e.g., choice of threshold and visual check of the obtained surface mesh). Nevertheless, this method assures an accurate representation of the investigated stent geometry, including the inhomogeneous deformations resulting from the crimping process. Furthermore, this approach may allow creating solid meshes in a semi-automated manner by sweeping the mesh along the centerline. Such a solid mesh may be required for very detailed stress analyses.

The triangulated surface depends on the threshold used during the segmentation process. Using a lower or higher threshold value may slightly modify the estimated strut cross-section. Using images with a voxel size of 10.6 μm, this means that the stent strut is approximately 20 μm smaller if 2 voxels are removed by using a different

Fig. 3 Comparison of the simulation results using two different methods to create the stent geometry. The simulation based on the traditional approach is depicted in **a**, whereas the beam mesh in **b** is based on the new approach involving the centerline computation



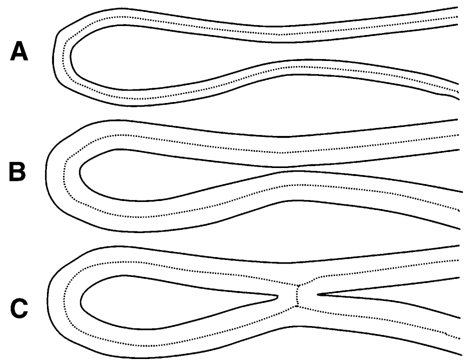


Fig. 4 **a**, **b** The same stent geometry using a different segmentation threshold. The resulting centerline in both cases is almost identical and therefore, the applied threshold has a limited impact on the resulting centerline. However, the centerline can be modified by the segmentation threshold in case that the reconstructed stent geometry contains connections between different struts which are not connected in reality (**c**)

threshold, which is important for a stent with struts of approximately 80 μm. However, the impact of the applied threshold on the resulting centerline will be small on condition that the surface mesh is not connected at locations where two separate struts come close to each other (Fig. 4). A second micro-CT scan was performed using a higher resolution (CT voxel size of 1.05 μm) in order to obtain a more accurate estimate of the strut cross-section (see Fig. 5), which is required for the definition of the beam elements. Using this high resolution, only a small portion of the stent could be scanned.

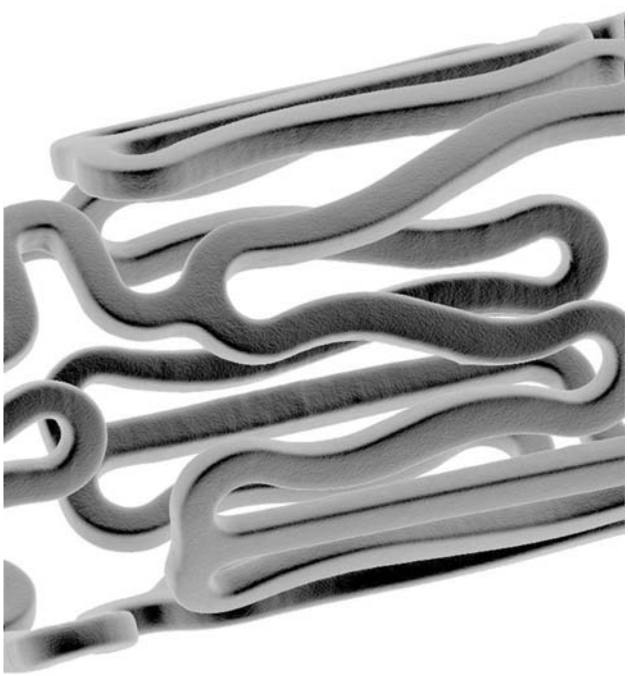


Fig. 5 A second micro-CT scan was performed using a higher resolution (CT voxel size of 1.05 μm) in order to obtain a more accurate estimate for the strut cross-section. Only a small portion of the stent was scanned at this high resolution

A limitation of the proposed approach is that the size of the elements of the finite element beam mesh is determined by the centerline algorithm and is independent of the curvature of the stent struts. However, it may be necessary to have more elements in regions of sharp geometry transitions.

Furthermore, it should be emphasized that the developed strategy is meant to create a finite element mesh of existing stent designs in order to compare their mechanical behavior. The method does not provide the possibility for parametric modeling, which is useful during the design phase of new devices [3].

4 Conclusion

During the last decade, numerical simulations have emerged as a valuable tool to investigate and develop medical devices, such as balloon-expandable stents. Studying the mechanical behavior of current stenting devices is sometimes time-consuming, partially because the different manual steps that occur during the preprocessing phase. In this study, we describe a new methodology to automate the process of generating a finite element beam mesh starting from a stent sample. The proposed strategy is time-effective and results in a beam mesh, accurately representing the investigated stent geometry.

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References

1. Antiga L (2002) Patient-specific modeling of geometry and blood flow in large arteries, PhD thesis. Politecnico Di Milano, Italy
2. Colombo A, Moses JW, Morice MC, Ludwig J, Holmes DR, Spanos V, Louvard Y, Desmedt B, Di Mario C, Leon MB (2004) Randomized study to evaluate sirolimus-eluting stents implanted at coronary bifurcation lesions. Circulation 109:1244–1249. doi: 10.1161/01.CIR.0000118474.71662.E3
3. De Beule M (2008) Finite element stent design, PhD thesis. Ghent University, Belgium
4. De Beule M, Mortier P, Carlier SG, Verhegghe B, Van Impe R, Verdonck P (2008) Realistic finite element-based stent design: the impact of balloon folding. J Biomech 41:383–389. doi: 10.1016/j.jbiomech.2007.08.014
5. Hall GJ, Kasper EP (2006) Comparison of element technologies for modeling stent expansion. J Biomech Eng 128:751–756. doi: 10.1115/1.2264382
6. Holzapfel GA, Stadler M, Gasser TC (2005) Towards a computational methodology for optimizing angioplasty treatments with stenting. In: Holzapfel GA, Ogden RW (eds) Mechanics of biological tissue. Springer, Heidelberg, pp 207–220

7. Kiousis DE, Gasser TC, Holzapfel GA (2007) A numerical model to study the interaction of vascular stents with human atherosclerotic lesions. Ann Biomed Eng 35:1857–1869. doi:10.1007/s10439-007-9357-z
8. Masschaele BC, Cnudde V, Dierick M, Jacobs P, Van Hoorebeke L, Vlassenbroeck J (2007) UGCT: new X-ray radiography and tomography facility. Nucl Instrum Methods A 580:266–269. doi:10.1016/j.nima.2007.05.099
9. Migliavacca F, Petrini L, Montanari V, Quagliana I, Auricchio F, Dubini G (2005) A predictive study of the mechanical behaviour of coronary stents by computer modelling. Med Eng Phys 27:13–18. doi:10.1016/j.medengphy.2004.08.012
10. Mortier P, De Beule M, Carlier SG, Van Impe R, Verhegghe B, Verdonck P (2008) Numerical study of the uniformity of balloon-expandable stent deployment. J Biomech Eng 130(021018):1–7
11. Lally C, Dolan F, Prendergast PJ (2005) Cardiovascular stent design and vessel stresses: a finite element analysis. J Biomech 38:1574–1581. doi:10.1016/j.jbiomech.2004.07.022
12. Radaelli AG, Augsburger L, Cebral JR, Ohta M, Rüfenacht DA, Balossino R, Benndorf G, Hose DR, Marzo A, Metcalfe R, Mortier P, Mut F, Reymond P, Socci L, Verhegghe B, Frangi AF (2008) Reproducibility of haemodynamical simulations in a subject-specific stented aneurysm model—a report on the virtual intracranial stenting challenge 2007. J Biomech 41:2069–2081. doi:10.1016/j.jbiomech.2008.04.035
13. Scheinert D, Scheinert S, Sax J, Piorkowski C, Bräunlich S, Ulrich M, Biamino G, Schmidt A (2005) Prevalence and clinical impact of stent fractures after femoropopliteal stenting. J Am Coll Cardiol 45:312–315. doi:10.1016/j.jacc.2004.11.026
14. Serruys PW, de Jaegere P, Kiemeneij F, Macaya C, Rutsch W, Heyndrickx G, Emanuelsson H, Marco J, Legrand V, Materne P, Belardi J, Sigwart U, Colombo A, Goy JJ, van den Heuvel P, Delcan J, Morel M (1994) A comparison of balloon-expandable-stent implantation with balloon angioplasty in patients with coronary artery disease. N Engl J Med 331:489–495. doi:10.1056/NEJM199408253310801